Topic 19
Review & Evaluation

Review and Evaluation in Software Project Management refers to the systematic process of analyzing the progress and outcome of a project. It helps identify issues, challenges, and successes, ensuring that corrective actions are taken as needed.

1. Project Reviews

A **project review** is a formal or informal evaluation at various stages of a project to assess its progress, performance, and alignment with the objectives. It is typically conducted by project managers, team members, stakeholders, or external parties.

Types of Project Reviews:

- **Formal Reviews:** These are planned meetings held at specific points in the project. They follow a structured agenda and are usually documented. Examples include:
  - **Design Review:** Reviewing the project's design to ensure it meets requirements.
  - **Code Review:** Evaluating the code for quality, standards, and functionality.
  - **Test Review:** Analyzing test cases, test results, and overall testing strategy.
- **Informal Reviews:** These are spontaneous discussions that may occur throughout the project to address concerns or improvements. They are more flexible and may not be documented as formal reviews.

Purpose of Project Reviews:

- **Tracking progress:** To monitor the status of tasks, deadlines, and resources.
- **Identifying risks:** To uncover potential issues early in the project.
- **Improving quality:** To ensure the project is meeting quality standards at each phase.
- **Stakeholder engagement:** To keep stakeholders informed and involved in the process.

Example:

- A project manager conducts a **Design Review** to ensure that the wireframes and prototypes align with the customer's expectations. The review might reveal that some features are missing, so adjustments are made before the development phase begins.

Key Components of Project Reviews:

1. **Objectives:** Clear goals that the review aims to achieve.
2. **Scope:** The boundaries of the review, such as which project deliverables are under examination.
3. **Participants:** The team members, stakeholders, and subject matter experts involved.
4. **Documentation:** A record of the review findings, decisions, and actions.
5. **Follow-up:** Plans for addressing issues and ensuring continuous improvement.

2. Post-Mortem Evaluation

A **post-mortem evaluation** (also known as a **retrospective**) is a process that takes place after the project has been completed or terminated. The purpose is to assess the entire project lifecycle, identify what went well, and pinpoint areas for improvement.

Objectives of Post-Mortem Evaluation:

- **Lessons Learned:** To identify best practices and mistakes that could influence future projects.
- **Process Improvement:** To understand how processes and practices can be refined for better results in the future.
- **Team Reflection:** To give the team a chance to discuss the project's challenges and successes.

Post-Mortem Process:

1. **Data Gathering:** Collect data related to the project's timeline, budget, quality, resources, and performance.
2. **Discussion:** Team members discuss their experiences, issues faced, and areas of improvement.
3. **Analysis:** Analyze the causes of issues and determine whether they were predictable or avoidable.
4. **Action Plan:** Develop strategies and action items to improve project management practices.

Example: After completing a software development project, a team conducts a post-mortem evaluation. They find that delays occurred due to lack of communication between development and QA teams. As a result, they propose an improved communication plan for future projects.

Benefits of Post-Mortem Evaluation:

- **Continuous Improvement:** Helps refine processes and practices over time.
- **Team Growth:** Builds trust and communication among team members.
- **Risk Management:** Helps in identifying potential risks for future projects.

Common Areas of Focus in Post-Mortem:

- **Timeline:** Did the project meet deadlines? If not, why?
- **Budget:** Was the project completed within the allocated budget? If not, where did overspending occur?
- **Quality:** Were there any quality issues, and how were they addressed?
- **Team Performance:** How well did the team collaborate? Were there any communication gaps?
- **Client Satisfaction:** Was the final product aligned with client expectations?

Scenario-Based Questions with Solutions:

Scenario 1: Question: During a project review, the client indicates that the user interface (UI) design does not match their expectations. The project team had already completed the initial design phase, and the client did not provide any feedback earlier. How should the project manager handle this situation?

Solution:

- **Step 1: Document the feedback**: The project manager should document the client's concerns about the UI.
- **Step 2: Evaluate the cause**: Investigate whether the client's expectations were clearly understood during the requirements gathering phase. If the feedback was not provided earlier, there may be a need to revisit communication practices.
- **Step 3: Assess impact**: Determine how significant the changes are and whether they will affect the project's timeline and budget.
- **Step 4: Adjust the design**: If feasible, update the design to meet the client's needs. Communicate the impact of the changes to the client regarding the timeline and costs.
- **Step 5: Follow-up review**: Set up a follow-up meeting with the client to review the changes and ensure alignment.

Scenario 2: Question: In a post-mortem evaluation, the development team realizes that communication gaps between the QA and development teams led to delays in delivering the product. What action should be taken to prevent this issue in future projects?

Solution:

- **Step 1: Acknowledge the issue**: During the post-mortem, openly discuss the communication gaps and their impact on the project.
- **Step 2: Identify root causes**: Determine whether the communication issues were due to a lack of formal processes or unclear roles between the teams.
- **Step 3: Implement solutions**:
  - Introduce regular check-ins or stand-up meetings between development and QA teams.
  - Use collaborative tools like Slack or Jira to ensure seamless communication and tracking.
  - Set up clear expectations for when and how QA should begin testing the code.
- **Step 4: Monitor progress**: Track the effectiveness of these changes in subsequent projects through periodic reviews.

Scenario 3: Question: A software project has gone through several formal reviews, but there have been multiple delays due to poor resource management. How can the project manager improve resource allocation in future projects?

Solution:

- **Step 1: Review the resource management process**: Analyze how resources were allocated in this project and identify inefficiencies or issues.
- **Step 2: Develop a resource plan**: In future projects, create a detailed resource management plan outlining the number of resources needed for each phase, along with their availability.

- **Step 3: Use project management tools**: Utilize tools like Microsoft Project or Asana to monitor resource allocation in real-time.
- **Step 4: Allocate buffer time**: Plan for buffer time to accommodate unexpected issues or delays.
- **Step 5: Continuous improvement**: Regularly assess resource allocation throughout the project to plan as necessary.
